of antimatter (the positron or antiproton or antineutron). When a particle of matter meets its corresponding antiparticle of antimatter, they annihilate each other.

apparent horizon. The outermost location around a black hole, where photons, trying to escape, get pulled inward by gravity. This is the same as the *(absolute) horizon* only when the hole is in a quiescent, unchanging state.

astronomer. A scientist who specializes in observing cosmic objects using telescopes.

astrophysicist. A physicist (usually a theoretical physicist) who specializes in using the laws of physics to try to understand how cosmic objects behave.

astrophysics. The branch of physics that deals with cosmic objects and the laws of physics that govern them.

atom. The basic building block of matter. Each atom consists of a nucleus with positive electric charge and a surrounding cloud of electrons with negative charge. Electric forces bind the electron cloud to the nucleus.

atomic bomb. A bomb whose explosive energy comes from a chain reaction of fissions of uranium-235 or plutonium-239 nuclei.

band. A range of frequencies.

bandwidth. The range of frequencies over which an instrument can detect a wave.

bar detector. A gravitational-wave detector in which the waves squeeze and stretch a large metal bar, and a sensor monitors the bar's vibrations.

beam splitter. A device used to split a light beam into two parts going in different directions, and to combine two light beams that come from different directions.

big bang. The explosion in which the Universe began.

big crunch. The final stage of recollapse of the Universe (assuming the Universe does ultimately recollapse; we don't know whether it will or not).

binary system. Two objects in orbit around each other; the objects may be stars or black holes or a star and a black hole.

BKL singularity. A singularity near which tidal gravity oscillates chaotically both in time and space. This is the type of singularity that